

Numerical Schemes

In this section, the transition from the continuous partial differential equation to discrete algebraic forms suitable for C++ implementation is presented.

To solve the one-dimensional transient diffusion equation, previously introduced in Theoretical Background as Equation (1):

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \quad (1)$$

the continuous temporal and spatial derivatives must be approximated using finite difference expressions. Implicit numerical schemes are therefore introduced to obtain discrete formulations of the governing equation.

Implicit Numerical Schemes

1. **Laasonen Scheme**
2. **CrankNicolson Scheme**